INSTRUCTIONS — SCR ELECTRIC VEHICLE CONTROL

IC4484 AUXILIARY PLUGGING CONTROL

E-701 (36-84 VOLTS) E-702 (84-144 VOLTS) RESISTOR AND NORMALLY OPEN CONTRACTORS

Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, DISCONNECT THE BATTERY AND DISCHARGE CAPACITOR 1C.

FUNCTION

The function of the auxiliary plugging control, Fig. 1, or "B" kit is to insert a resistor in series with the motor armature during a "plugging" operation. This resistor dissipates most of the energy used in stopping the vehicle and forces the field current to approach the armature current thus reducing armature current and improving commutation.

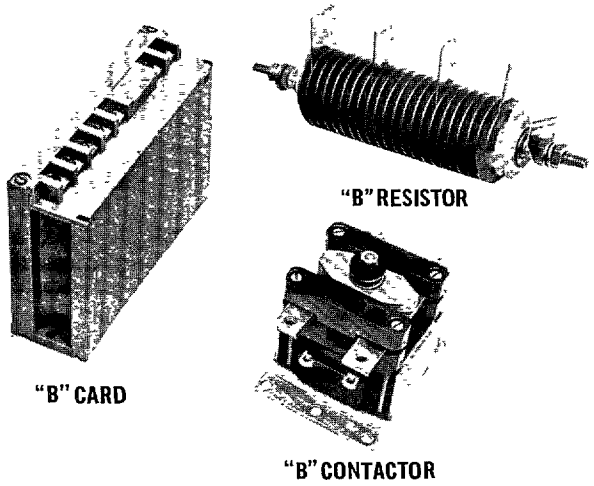


Fig. 1. IC4484 Auxiliary plugging control

OPERATION

The auxiliary plug card is connected in the EV-1 SCR circuit as shown in Fig. 2. The logic in the "B" card energizes the "B" coil when the main SCR is pulsing and no "plug" signal occurs at R10. During normal running, the "B" contactor is closed and the "B" resistor is by-passed.

When the truck is plugged, a "plug" signal occurs at R10 which de-energizes the "B" coil and a negative signal is produced at A1 by the motor armature. The logic in the "B" card clamps R7 to zero. Near the end of the plug, the negative signal at A1 becomes zero and the logic in the "B" card puts a positive voltage into R7 (as adjusted by the trimpot on the "B" card). CW reduces the speed in the new direction needed to cancel the plug signal which re-energizes the "B" coil.

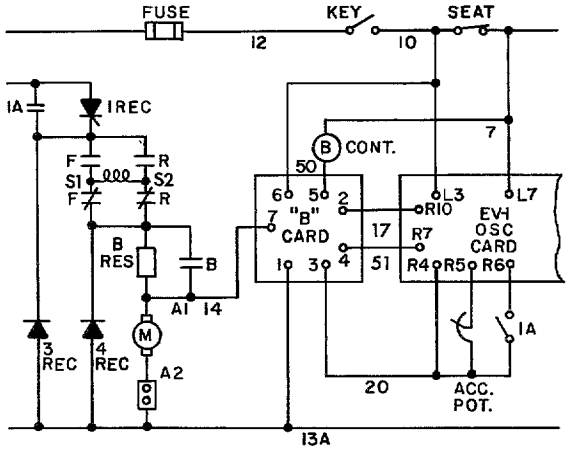


Fig. 2. EV-1 SCR circuit showing connection of auxiliary plug card

The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

INDUSTRIAL CONTROL DEPARTMENT, GENERAL ELECTRIC COMPANY, U.S.A.

11-76

PRINTED IN U.S.A.